



DS&AI LAB PROJECT • TERM MAY 2026

Course-Aware Personalized Learning Companion for IIT Madras BS Degree Students

MILESTONE 2: Dataset Identification & Preparation

Indian Institute of Technology Madras | Group 3

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Introduction & Objectives

Milestone 2 identifies, evaluates, and prepares the knowledge base powering a Retrieval-Augmented Generation (RAG) assistant for CS2007, Machine Learning Techniques (MLT). The system answers conceptual doubts, PYQ-style questions, and weekly content clarifications by retrieving grounded context and generating natural-language answers via an LLM (Gemini/Groq).

1

Document Ingestion & Chunking

Converting heterogeneous course resources into retrieval-ready text chunks.

2

Hybrid Retrieval

Combining Dense Retrieval (BGE-small embeddings) and Sparse Retrieval (BM25 exact keyword matching) using Qdrant's native Hybrid Search with Reciprocal Rank Fusion (RRF).

3

Answer Generation

Conditioning an LLM (Gemini/Groq) on retrieved context to produce a grounded, course-accurate response.

The knowledge base must be: clearly sourced with verified ownership, well-structured and described, assessed for quality issues, adequate in coverage with augmentation plans, split without leakage, and processed through a documented, reproducible pipeline.

SCOPE

In-scope: CS2007 weekly text content (Transcripts, Notes, PYQ, AQ/PQ, External Notes, FAQ), hybrid retrieval, fusion/reranking.

Out-of-scope: Other courses, video retrieval, live forum threads, LLM fine-tuning.

SECTION 2

Data Source Identification & Verification

Source	Description	Owner/Platform	Format
Discourse Weekly Resources Thread	Official weekly resource index	IITM Discourse	HTML, attachments
Karthik Sir Notes	Instructor-maintained structured weekly notes	Course faculty (GitHub Pages)	Static HTML/Markdown
Transcripts	Lecture video transcripts, week-wise	Course platform / instructor	Text (VTT/plain)
PYQ	Past end-term/quiz question papers	IITM Academic Office	PDF
AQ/PQ (week-wise)	Weekly graded and practice questions	Course platform	HTML/PDF
External Notes	Community/TA-contributed notes	Student contributors	Markdown/PDF
FAQ	Curated doubts and TA responses	Discourse threads	HTML

Ownership: All content originates from IITM's official course platforms, accessed as an enrolled student for internal academic use only, not redistribution. PYQs are institute-released archives used strictly for retrieval context. Any open-source augmentation material will be cited with source and license.

SECTION 3

Dataset Description

Composition by Content Type (94 docs, 1.25 MB)

- **Lecture Transcripts:**
38 files, Markdown, converted from PDFs; timestamps preserved
- **Instructor Notes:**
12 files, Markdown, cleaned from raw GitHub scraping
- **PYQ:**
8 files, Markdown, OCR extracted (EasyOCR) for math text
- **AQ/PQ (week-wise):**
12 files, Markdown, OCR extracted
- **External Notes:**
12 files, Markdown, converted from PDF
- **FAQ:**
12 files, Markdown, scraped from mlt.pulki.in

METADATA SCHEMA (per chunk)

doc_id / chunk_id

Unique identifiers for citation and traceability

source_type

notes | transcript | pyq | aq_pq | external | faq

week / lecture_number / page_number

Populated per source type; null when not relevant

title / url / author

Provenance for retrieval-time filtering and display

chunk_text / chunk_index / total_chunks_in_doc

Chunk content and position within source document

token_count / language

Used for chunk-size tuning and language checks

confidence_flag

verified vs low_confidence, for unverified FAQ/external content

Data Quality Assessment

Missing Values

Status: Minor coverage gaps observed in early weeks for External Notes

Mitigation: Documented as coverage gaps rather than imputed; flagged for augmentation

Duplicates

Status: Zero exact duplicates found within the verified dataset

Mitigation: No merging required; all documents ingested natively

Noise

Status: Heavy filler words, timestamps, PyMuPDF artifact tags in raw transcripts

Mitigation: Regex-based cleanup; timestamps converted to Markdown headers

Formatting Inconsistency

Status: Notes mixed Markdown, LaTeX, inline HTML; PYQs were scanned PDFs

Mitigation: Unified plain text; LaTeX preserved; math extracted via EasyOCR

Broken/Stale Links

Status: Zero 404s encountered during local corpus processing

Mitigation: N/A; all ingestion performed locally on verified files

FAQ Answer Correctness

Status: FAQ answers from mlt.pulki.in are community/TA replies

Mitigation: Unverified entries tagged with source metadata for LLM citation

Adequacy, Augmentation & Split Strategy

Coverage Gaps (Expected)

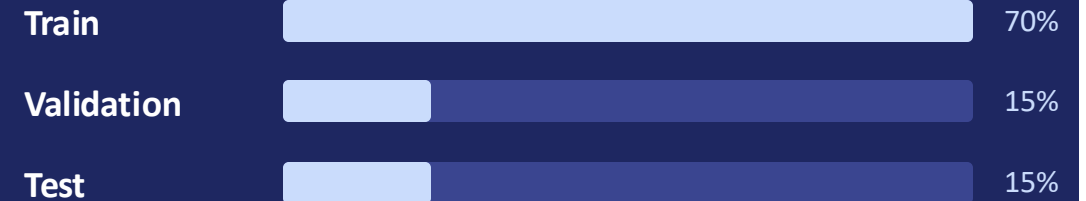
Weeks with sparse External Notes or no PYQ history may leave thin retrieval context. FAQ coverage is expected to skew toward assignment-deadline doubts rather than conceptual depth. Both to be confirmed with per-week counts.

Augmentation Plan

- Supplementary retrieval-only sources (instructor slides, cited open textbook sections) for thin weeks
- Synthetic Q&A (approx. 5-10 per week) for retrieval evaluation only, never for the index
- LLM-generated questions grounded in a source chunk, stored as gold passage
- Manual review of a sample for relevance and naturalness before use

TRAIN / VALIDATION / TEST SPLIT

Split by week/topic, not randomly by row, to prevent leakage between related chunks and queries.



Proposed ratio; may adjust once actual week/query counts are known. Leakage check verifies no gold-context chunk appears (verbatim or near-duplicate) in the training-time index.

Generation eval: a held-out set of about 50 queries (approx. 4/week) across all 12 weeks, reserved purely for end-to-end answer-quality evaluation.

RAG Pipeline: Chunking, Embedding & Vector Store

Chunking Strategy

Chunk size: 384 tokens (fits BGE-small's 512-token limit)

Overlap: 15% (approx. 58 tokens), prevents orphaned sentences

Strategy: Semantic/Recursive via LangChain

RecursiveCharacterTextSplitter

Special case: FAQ/PYQ/AQ-PQ kept as one atomic Question+Solution chunk

Embedding Model: BGE-small

384-dim output, open-weight, no API cost or rate limits.

Chosen over: OpenAI text-embedding-3-small (paid API dependency), e5-small-v2 and gte-small (comparable but less community adoption / slightly weaker MTEB scores).

Rationale: no external dependency, small footprint for CPU/local dev, strong MTEB retrieval performance out of the box.

Vector Database Decision: Qdrant

Selected over FAISS for native payload filtering. Given the extensive metadata schema, the ability to filter by week and source_type before top-k retrieval is critical, e.g. restricting a query to week 4 PYQs only. Qdrant's local Python client also allows file-based persistence without a Docker server during development.

	FAISS	Qdrant
Deployment		In-process only Server or local file
Metadata filter		Manual Native payload filter
Persistence		Manual save/load Built-in

Evaluation Metrics & Hybrid Search

Retrieval Evaluation (k = 5, secondary k = 10)

Recall@5

Fraction of queries where the gold chunk appears in top-5 retrieved

Precision@5

Fraction of top-5 retrieved chunks relevant to the query

MRR

Rewards ranking the gold chunk higher within top-k

Generation Evaluation (RAGAS or equivalent LLM-judge)

Faithfulness

Whether the generated answer is supported by retrieved context, no hallucinated claims

Answer Relevance

Whether the generated answer actually addresses the student's question

Context Precision/Recall

Whether retrieved context was necessary and sufficient for the answer

Hybrid Search: Dense + Sparse via Reciprocal Rank Fusion

Rather than a slow cross-encoder reranker, Qdrant's native hybrid search combines Dense Search (semantic meaning via BGE-small) with Sparse Search (exact keyword matching via BM25), fused with Reciprocal Rank Fusion. In a technical course like MLT, exact keyword matches such as variable names or acronyms (PCA) matter as much as semantic meaning, giving reranker-level precision without cross-encoder latency.

SECTION 8

Preprocessing Pipeline (Reproducibility)



Scrape:	Discourse + karthik-iitm.github.io/MLT ; raw HTML/PDF/Markdown saved with source URL retained.
Parse & Clean:	PyMuPDF4LLM for PDFs with EasyOCR fallback for scanned math, BeautifulSoup for HTML, direct Markdown parse. Strip boilerplate, timestamps, filler words; normalize LaTeX.
Chunk & Tag Metadata:	LangChain RecursiveCharacterTextSplitter (384 tokens / 15% overlap) with MarkdownHeaderTextSplitter; atomic Q&A chunks; tagged with Course, Week, Lecture, Timestamp, Source.
Embed, Index, Retrieve & Evaluate:	BGE-small dense + BM25 sparse vectors (FastEmbed ONNX) indexed in Qdrant; Hybrid Retrieval (RRF) feeds Gemini for cited answers; topic-wise split; Recall@k, Precision@k, MRR on held-out queries.

All scripts are version-controlled on the group GitHub repository, with raw scraped data, cleaned text, and final chunk/embedding artifacts stored in separate pipeline stages for full reproducibility.

TEAM & INDIVIDUAL CONTRIBUTIONS

Group 3 | Milestone 2 Consent & Contribution Record

Mayank Singh

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Defined the dataset scope and objectives; identified official data sources, ownership constraints, and the overall dataset description and metadata schema.

Ali Jawad

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Designed the RAG data pipeline: document preprocessing, LangChain orchestration, chunking strategy, embedding model (BGE-small), vector database selection (Qdrant), hybrid retrieval, and evaluation metrics.

Sachi Dhaturaha

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Conducted the data quality assessment, analyzed dataset adequacy and coverage gaps, and prepared the augmentation strategy, train/validation/test split methodology, and leakage prevention plan.

Aryan Pratap Maurya

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Developed the reproducible preprocessing workflow: document parsing, cleaning, metadata tagging, plus the end-to-end pipeline architecture and Mermaid workflow diagram.

Jibin V Mathews

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Reviewed and consolidated all sections into a consistent report, refined technical writing and formatting, verified cross-section consistency, compiled references, and finalized the submission.